

Formal Verification Supplement: Infinite Quadratic Aggregation

1 Formal verification of the infinite-aggregation example

The third Lean package concerns the explicit family

$$S_r = \{(u, v) \in \mathbb{R}^r \times \mathbb{R}^r : \|u\|^2 < 1, \|v\|^2 < 1, u^\top v > 1/2\}, \quad r \geq 2.$$

Squared lengths and inner products in the formalization are Euclidean coordinate sums. The package treats the actual quadratic map and ordinary convex hull, including the four-variable case $r = 2$.

Geometry and good multipliers. The set S_r is nonempty and bounded, and its ordinary convex hull is proper. Its homogeneous map is

$$(u, v, t) \mapsto (\|u\|^2 - t^2, \|v\|^2 - t^2, t^2/2 - u^\top v).$$

The formal HHC assertion requires convexity of this image on the kernel of every nonzero real linear functional. It is proved for all $r \geq 2$; HHC is not supplied as an additional premise for this example.

The proof uses an explicit orthonormal frame and an algebraic unit-circle construction to obtain the full range of a linear functional on a fixed two-column Gram fiber. For a hyperplane $\text{tr}(P^\top U) + st = 0$, put $G = U^\top U$, $H = P^\top P$, and $T = t^2$. Its exact feasible Gram domain is

$$G \succeq 0, \quad T \geq 0, \quad s^2 T \leq \text{tr}(HG) + 2\sqrt{\det H \det G}.$$

Both necessity and realization are proved, including singular Gram matrices and zero coefficients. Concavity of the square root of the two-by-two determinant proves convexity of this domain. The homogeneous quadratic image is its linear image, which gives HHC.

A good multiplier is defined by the original conditions: it is nonzero and nonnegative, its homogeneous aggregate has at most one negative eigenvalue counted with multiplicity, and its strict aggregation contains $\text{conv } S_r$. The proved classification is

$$\lambda \text{ is good} \iff \lambda \geq 0, \quad \lambda \neq 0, \quad \lambda_3^2 \leq 4\lambda_1\lambda_2.$$

The homogeneous matrix consists of r copies of

$$B_\lambda = \begin{pmatrix} \lambda_1 & -\lambda_3/2 \\ -\lambda_3/2 & \lambda_2 \end{pmatrix}$$

and the scalar $\lambda_3/2 - \lambda_1 - \lambda_2$, up to coordinate order. For nonnegative multipliers violating the discriminant inequality, the repeated block gives at least r negative eigenvalues. For nonzero nonnegative multipliers satisfying it, the block is PSD and the scalar is strictly negative, so there is exactly one negative eigenvalue. Convexity of the original aggregate and its strict negativity on S_r establish strict negativity throughout the ordinary convex hull. Thus the classification does not assume a general aggregation hull-description theorem.

Indispensable strict rays. For every $\tau \in [1, 2]$, the package constructs actual vectors whose Gram matrix is

$$\begin{pmatrix} 1 - 1/(10\tau) & 2/5 \\ 2/5 & 1 - \tau/10 \end{pmatrix}.$$

Their point x_τ has residual vector $\frac{1}{10}(-1/\tau, -\tau, 1)$. Every good aggregate is nonpositive there, with equality exactly for positive multiples of $\rho(\tau) = (\tau, 1/\tau, 2)$. The multiplier $\rho(\tau)$ is itself good, so $x_\tau \notin \text{conv } S_r$; every good multiplier on a different ray is strictly satisfied at this point.

Consequently, if a family of strict good aggregations describes $\text{conv } S_r$ exactly, it contains a positive multiple of $\rho(\tau)$ for every $\tau \in [1, 2]$. The formal cardinality argument normalizes weights to third coordinate two and proves that the set of normalized weights is uncountable. This counts distinct rays, not repeated indices in a family. In particular, no countable family of strict good aggregations describes the ordinary hull. This statement is conditional on an exact description; the package does not separately prove that the intersection of all good strict aggregations equals the hull.

The closed hull requires a different witness. For any finite family of good multipliers, choose an omitted ray and decrease the Gram off-diagonal $2/5$ by a sufficiently small positive amount. The perturbed Gram matrix remains positive definite and is realized by actual vectors. Every selected aggregate is strictly negative there, while the omitted aggregate is strictly positive. Every good aggregate is nonpositive on $\text{cl}(\text{conv } S_r)$ by continuity. Hence the perturbed point is outside this closed hull, although it satisfies every selected weak aggregation. No finite intersection of weak good aggregations can therefore equal $\text{cl}(\text{conv } S_r)$.

Scope and records. The Lean sources are in `Formal.InfiniteAggregation`; the frozen claims, declaration coverage, independent reviews, and targeted verification records are in `formal/topics/29-infinite-aggregation/`. This package does not establish an exact hull formula, countable weak aggregation sufficiency, equality with the hull of the original weak system, a finite SDP lift, an obstruction to arbitrary quadratic descriptions, approximation bounds, or the general sharp Gram-map theorem. It also makes no literature-priority claim.